Project 2: Mouse gastrulation scRNA-seq (E6.5-E7.5)

Decision log and limitations. Dataset: Pijuan-Sala et al. 2019 atlas (E-MTAB-6967), analysed with Scanpy and scVI.

1. Data and design

- Files used: the authors' processed matrix and metadata (not the FASTQs deposited at ArrayExpress). Full file: 29,452 genes x 139,331 cells.
- Of the 139,331 cells, 116,312 carry an authors' label (the published atlas size). The authors flagged 9,466 as stripped nuclei and 13,556 as doublets.
- Subset analysed: E6.5, E7.0, E7.5 = 15 samples, 33,144 cells. Excluded: E6.75, E7.25 and the mixed_gastrulation samples.
- Replication: samples sharing a pool index (14/15 = pool 12; 30/31/32 = pool 23; 19/20 = pool 16) are probably repeat runs of one pool. This is inferred from the metadata and unconfirmed. The independent unit is therefore the pool, not the sample. Sample 11 is absent from the file for an unknown reason.
- Design risk: batch 3 contains only E7.0 cells in this subset (68% of E7.0), so stage, batch and pool are confounded there.

2. Decisions

1. Raw integer counts preserved (file 01 and `layers["counts"]`); every later object derives from them.
2. The authors' labels, QC flags and embeddings were held out until the comparison step; our annotation (`annot_v2`) was frozen before they were opened.
3. Gene index = Ensembl IDs. 78 genes share 37 symbols, so marker lookups check for a unique ID.
4. Mitochondrial fraction from the 13 protein-coding mt- genes.
5. The data were already filtered (minimum 5,000 UMIs, maximum mito 2.37%), so no low-quality tail was visible. No cells were removed on counts; per-sample 3-MAD flags (3,382 cells, 10.2%) were kept as columns only.
6. Scrublet per sample, expected rate 0.05 (untuned): 189 cells called, scores kept as columns.

7. Normalisation: total 10,000 per cell, then log1p.
8. HVG-2000 (Seurat flavour, batch_key = sequencing.batch), 50 PCs computed, 30 used, 15 neighbours.
9. Leiden at resolutions 0.3, 0.6, 1.0, 1.5 (arbitrary starting values).
10. scVI: 2,000 HVGs on raw counts, batch_key = sequencing.batch, 30 latent dimensions, NB likelihood, 100 epochs. Early stopping did not trigger, so convergence is not proven.
11. Broader gene set: 6,000 HVGs. The first run silently reused the 2,000-gene flag and was invalid; it was redone as v2 and the earlier conclusion was retracted.
12. Trajectory on embryonic groups only (extraembryonic groups excluded); PAGA on annotation groups; DPT with the root chosen among E6.5 epiblast cells of samples 1 and 5.
13. Sensitivity runs: without samples 14/15/18; without authors-excluded cells.

3. Findings

- Samples 14, 15 and 18 (median 6.9k-9.2k UMIs vs 13k-22k elsewhere) form separate clusters that persist in HVG-2000, HVG-6000 and scVI. Cluster composition tracks sample and depth, but stage, pool and sample are confounded, so the cause cannot be isolated.
- Marker-based annotation matched the authors' labels well for extraembryonic ectoderm, streak/nascent mesoderm, epiblast, mesendoderm and visceral/ExE endoderm.
- Annotation errors found only after the comparison: the endoderm candidate group was parietal endoderm; the posterior epiblast/streak group was 40% rostral neurectoderm; one cluster labelled ExE ectoderm was 97% stripped nuclei; two clusters were about 91% authors-flagged doublets.
- Of the 33,144 cells, 3,917 (11.8%) are authors-excluded (1,832 stripped, 2,085 doublets) and were included in the main analyses. Scrublet recovered 159 of its 189 calls among the authors' doublets but missed 1,926 of their 2,085.
- Agreement with authors' labels at about 17 clusters (ARI / NMI): HVG-2000 0.56 / 0.72; scVI 0.58 / 0.69; HVG-6000 (22 clusters) 0.50 / 0.69. Without samples 14/15/18: 0.64 / 0.74, 0.66 / 0.70, 0.62 / 0.73. Resolution mattered more than representation (HVG-2000 ARI 0.63 at 14 clusters, 0.40 at 35).

- Pseudotime: the order epiblast < posterior epiblast/streak < streak < mesendoderm < mesoderm held in all runs, including without the suspect samples and without authors-excluded cells. Stage medians in the main run were 0.029, 0.095 and 0.232 (E6.5, E7.0, E7.5), with E7.0 and E7.5 overlapping.

4. Real time vs pseudotime

Real time is the collection stage and is shared by all cells of a sample. Pseudotime is a distance from a chosen root cell along the neighbour graph, has no time unit, and spreads cells of one stage across a wide range (E7.0 contains epiblast and streak/mesoderm cells together). Here the root was chosen from E6.5 cells, so agreement with stage is partly circular. The within-epiblast E6.5 to E7.0 step shrank when samples 14/15/18 were removed.

5. Limitations

- Pool-level replication only; the number of embryos per pool is not known from the data.
- Stage, batch, pool and sample are partly confounded (batch 3 = E7.0 only; sample 18 = the only batch-2 E6.5 sample), so batch correction can remove biology and cannot be fully validated.
- Pre-filtered data: our QC was a second pass and could not assess the low-quality tail.
- Our own QC missed most authors-flagged stripped nuclei and doublets; the main analyses were run with them present.
- Definitive endoderm is under-sampled; the posterior epiblast/streak group mixes lineages, so the middle step of the trajectory is not a clean stage.
- E6.75 and E7.25 were excluded, leaving 0.25-0.5 day gaps; two stages (E6.5, E7.0) rest on few samples after sensitivity filtering.
- All cluster numbers, resolutions, HVG counts and neighbour settings were chosen by convention, not optimised. scVI did not reach early stopping.
- The authors' labels are an independent annotation, not ground truth. ARI and NMI reward matching their granularity.
- p-values from cell-level tests are inflated (cells from a few pools are not independent) and were not used.

- Alternative methods not tried: batch-aware integration with sample-level covariates, RNA velocity, a lineage-focused pseudotime that separates the neurectoderm cells, and cell-cycle regression.

6. Errors corrected during the work

- First 6,000-gene run was invalid (reused the 2,000-gene flag); redone.
- An early claim that the authors had already removed doublets was wrong for this file.
- Cluster 2 was provisionally read as posterior epiblast/streak; the authors' labels indicate rostral neurectoderm.